

Correlation

A mutual relationship or connection between two things. Example: Price and demand.

- **Correlation** is a statistical tool that helps to measure and analyze the degree of relationship between two variables.
- Correlation analysis deals with the association between two or more variables.
- The degree of relationship is expressed by coefficient which range from correlation ($-1 \leq r \leq +1$)
- According to Ya Lun Chou, “Correlation analysis attempts to determine the degree of relationship between variables”
- According to W I KING “Correlation means that between two series or groups of data there exist some causal relationship”
- Change in one value is reflected on the other.

TYPES OF CORRELATION

1. Positive Correlation:

- **Definition:** In a positive correlation, as one variable increases, the other variable also increases. Similarly, as one variable decreases, the other also decreases.

2. Negative Correlation:

- **Definition:** In a negative correlation, as one variable increases, the other variable decreases, and vice versa.

3. No Correlation:

- **Definition:** In a no correlation scenario, there is no consistent relationship between the variables. Changes in one variable do not predict changes in the other.

Examples

1. Positive Correlation:

- **Hours of Exercise and Fitness Level:** Generally, the more time you spend exercising, the better your fitness level.
- **Education Level and Income:** Higher levels of education tend to be associated with higher income.

2. Negative Correlation:

- **Number of Cigarettes Smoked and Lung Capacity:** As the number of cigarettes smoked increases, lung capacity tends to decrease.
- **Speed of Driving and Travel Time:** The faster you drive, the less time it takes to reach your destination.

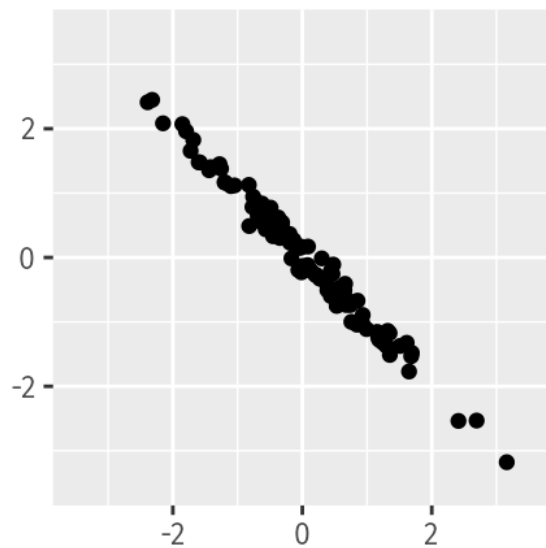
3. No Correlation:

- **Shoe Size and Intelligence:** There is no relationship between the size of your shoes and your intelligence.
- **Amount of Coffee Drunk and Color of Car:** The amount of coffee someone drinks has no relation to the color of their car.

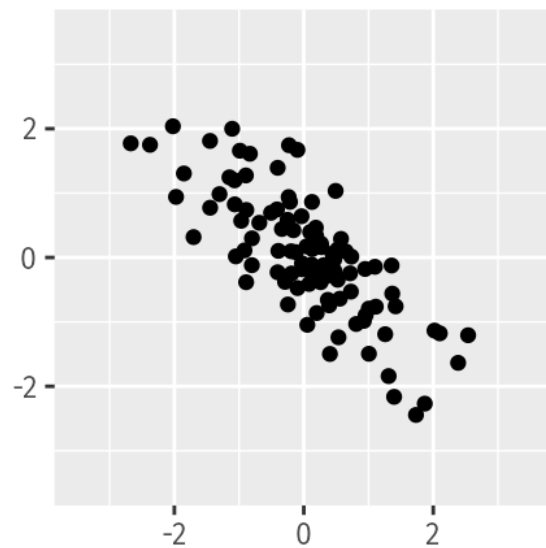
Scatter Diagram Method

- Scatter Diagram is a graph of observed plotted points where each point represents the values of X & Y as a coordinate. It portrays the relationship between these two variables graphically.

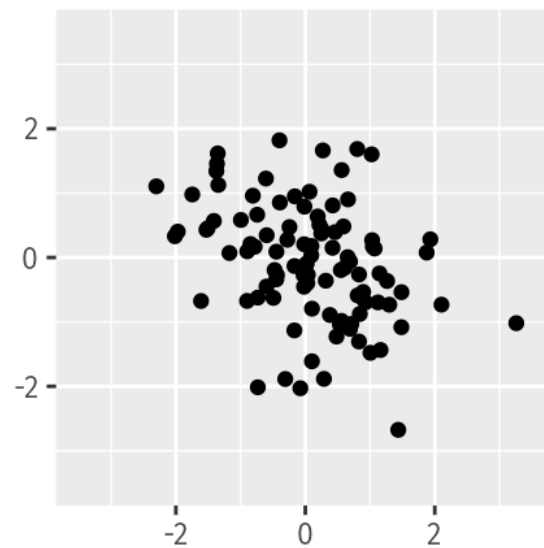
Correlation = -0.99



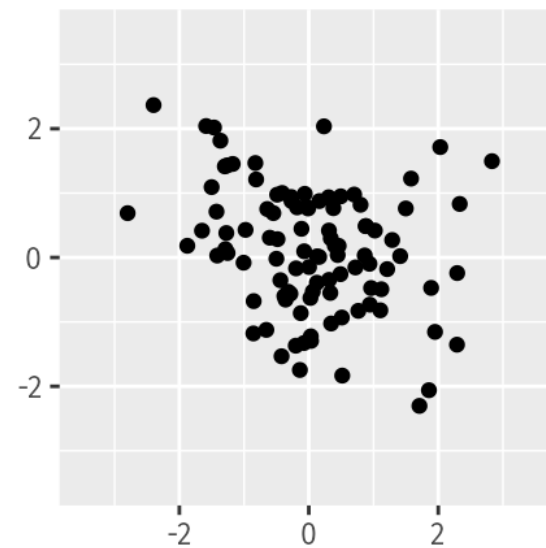
Correlation = -0.75



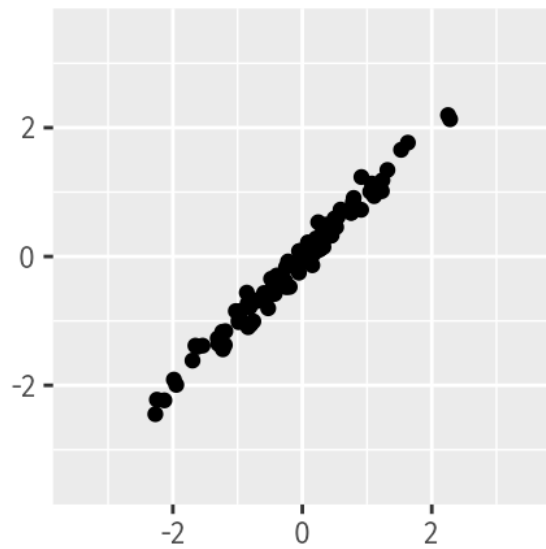
Correlation = -0.50



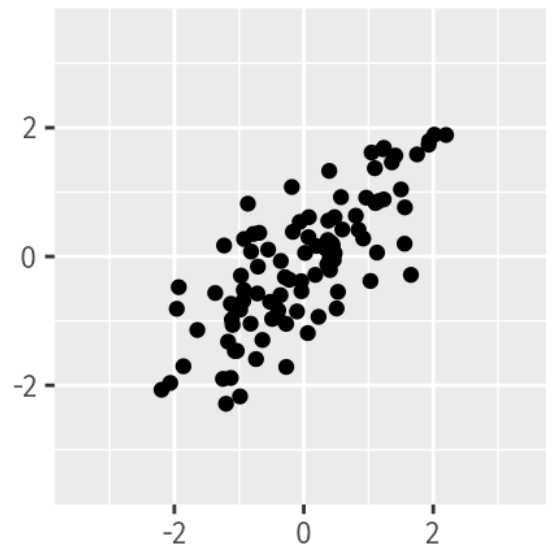
Correlation = -0.25



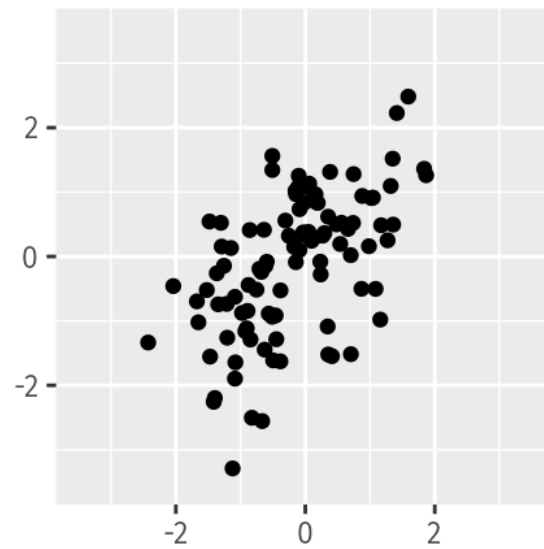
Correlation = 0.99



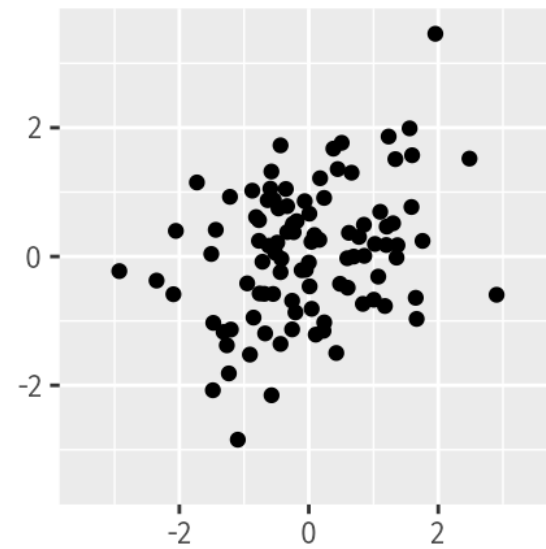
Correlation = 0.75



Correlation = 0.50

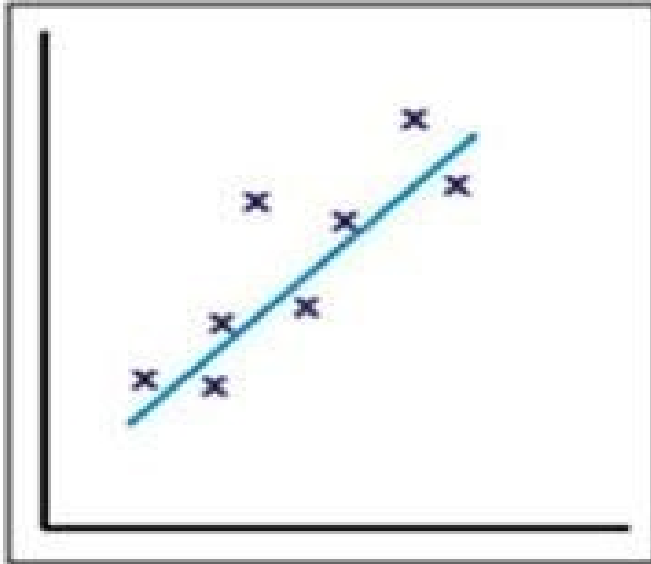


Correlation = 0.25



Types of Correlation

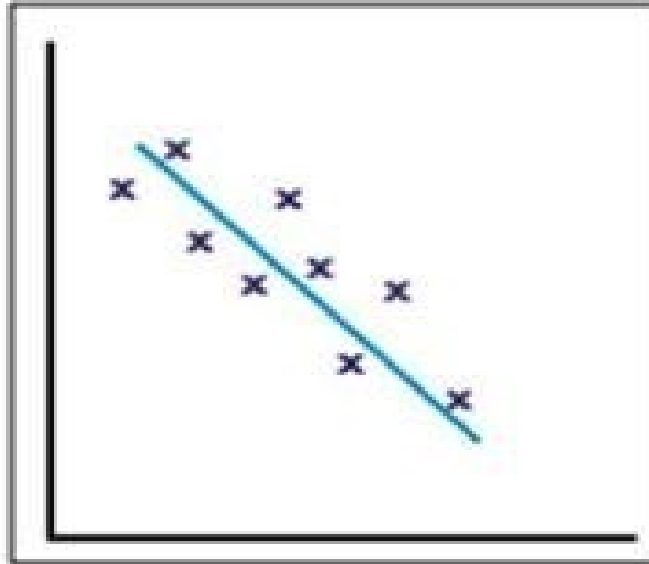
Positive correlation



The points lie close to a straight line, which has a positive gradient.

This shows that as one variable **increases** the other **increases**.

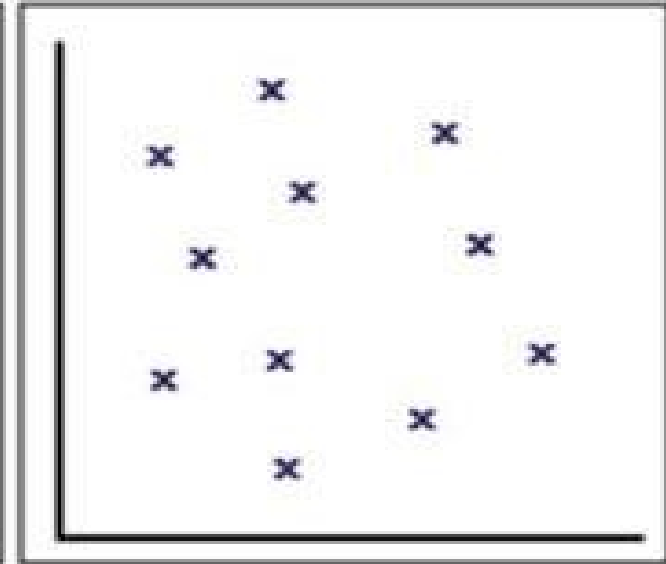
Negative correlation



The points lie close to a straight line, which has a negative gradient.

This shows that as one variable **increases**, the other **decreases**.

No correlation



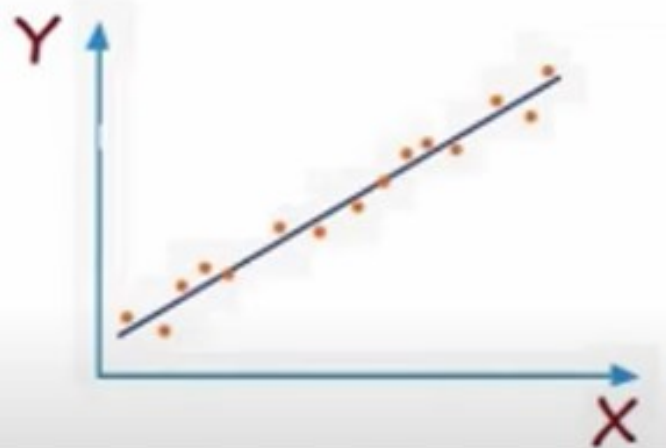
There is no pattern to the points.

This shows that there is **no connection** between the two variables.

Positive Correlation: If the values of the two variables deviate (moves) in the same direction, i.e., if the increase (decrease) in the values of one variable results, on an average, in a corresponding increase (decrease) in the values of the other variables

Examples:

- 1) Height and weights.
- 2) Family income and expenditure on luxury items.
- 3) Price and supply of a commodity



X	2	5	8	10
Y	18	25	34	51

X increases; Y also increases ----- Positive Correlation

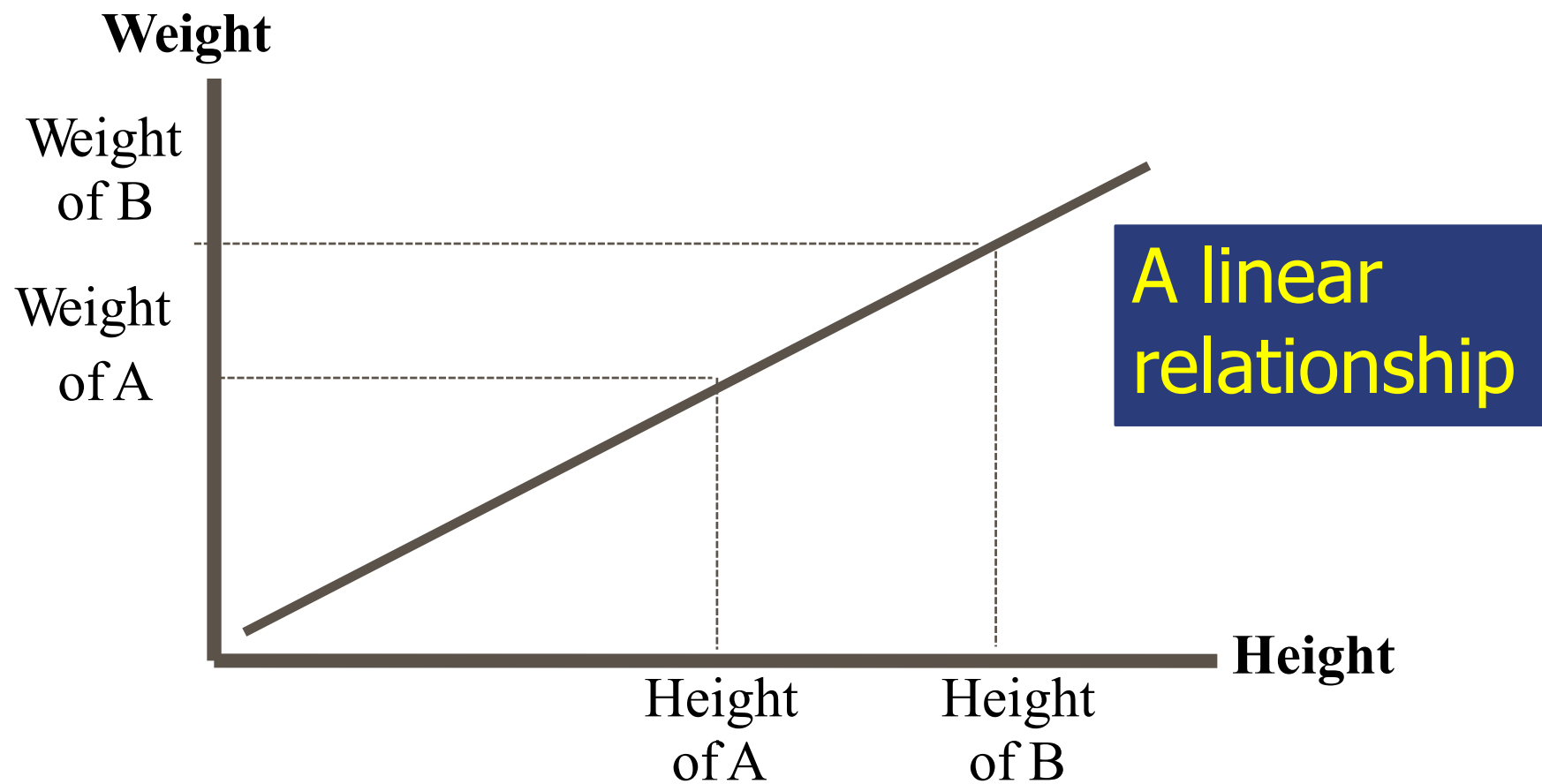
The **degree of correlation** refers to the strength and direction of the relationship between two variables. It is typically measured using the correlation coefficient, denoted as r , which ranges from -1 to $+1$.

Perfect Positive Correlation ($r = +1$): All data points lie exactly on a straight line (linear relationship) that slopes upward. This means that for every unit increase in one variable, the other variable increases by a fixed proportion.

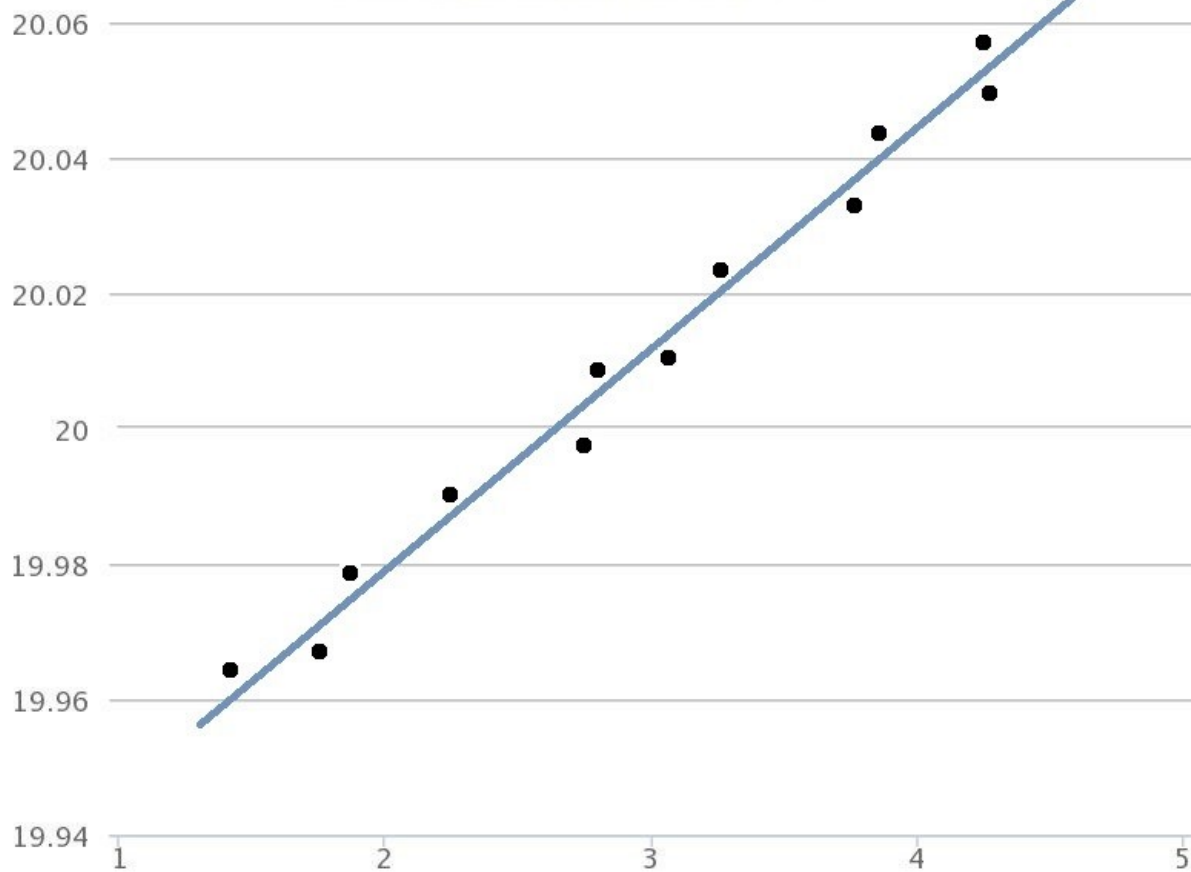
High Degree of Positive Correlation (Strong positive) (r close to $+0.7$ to $+0.99$): The data points are close to forming a straight line that slopes upward, **but there may be some slight deviations**. As one variable increases, the other tends to increase as well, but not exactly in the same proportion. This indicates a strong relationship between the variables, but not a perfect one.

Moderate Positive Correlation (r around +0.4 to +0.59): The data points still tend to slope upward, **but they are more spread out from the line**. The relationship is still positive but less strong.

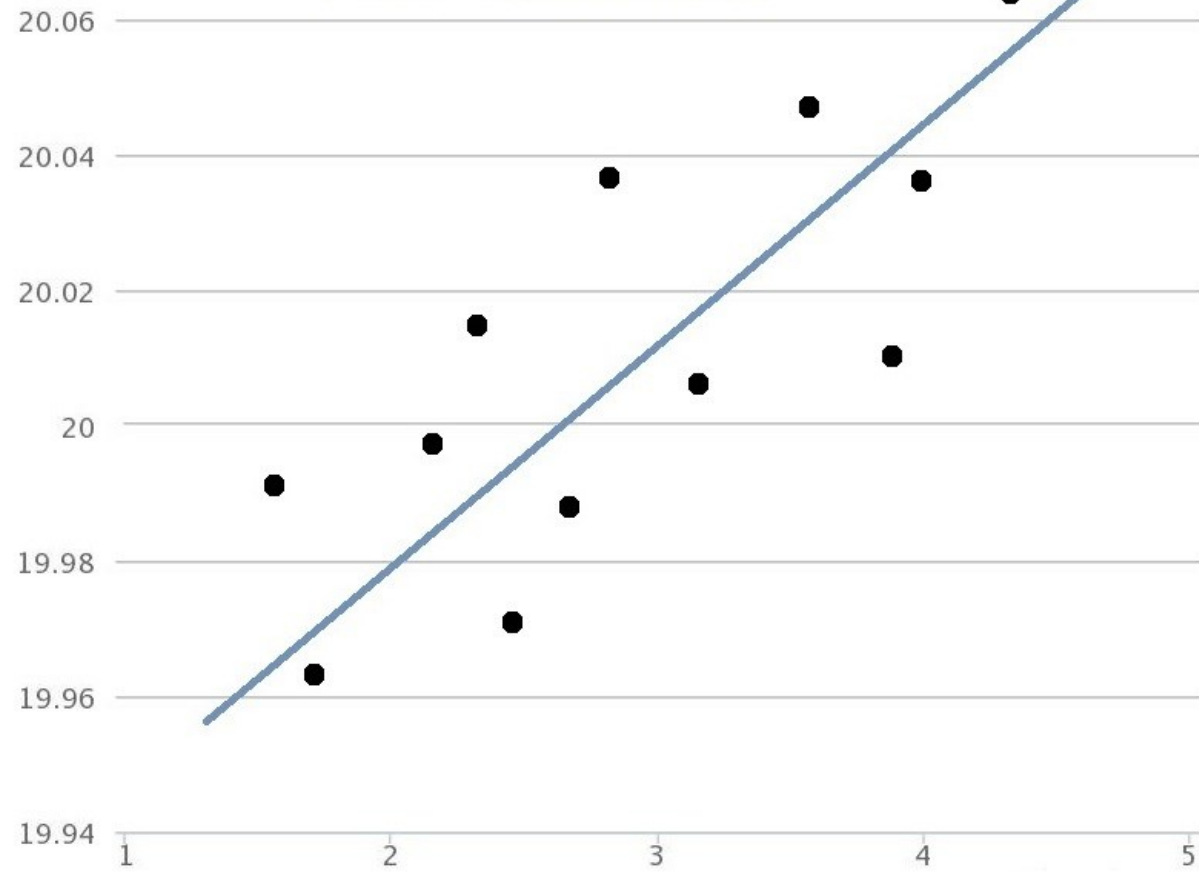
Weak Positive Correlation (r around +0.2 to +0.39): The data points show an upward trend, but the relationship is weak, meaning **the points are widely scattered**, and the correlation is not strong.

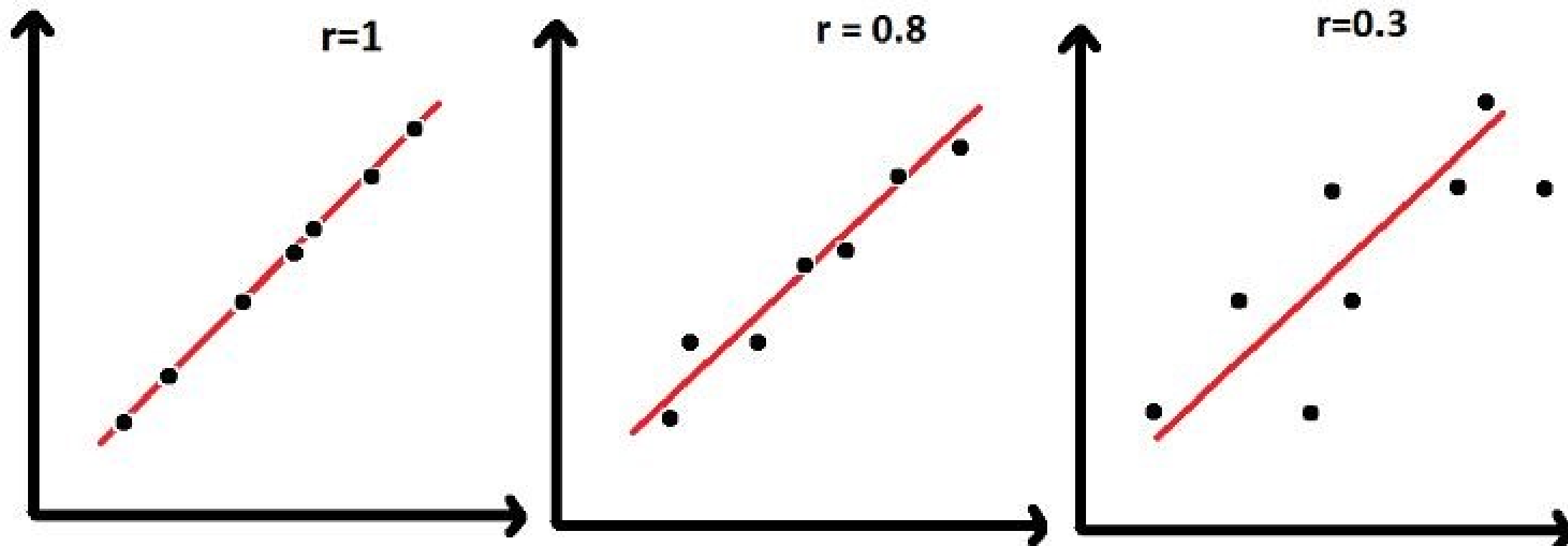


Strong Positive Correlation



Weak Positive Correlation



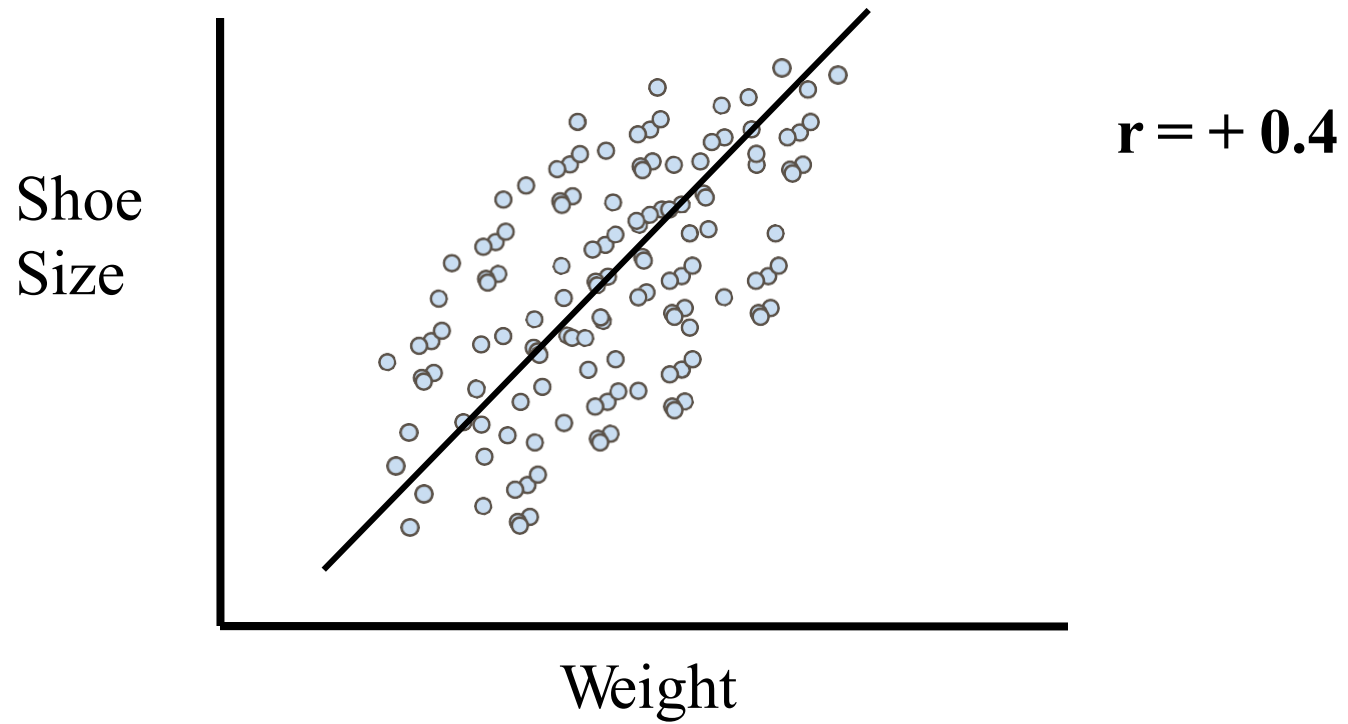


Perfect Positive

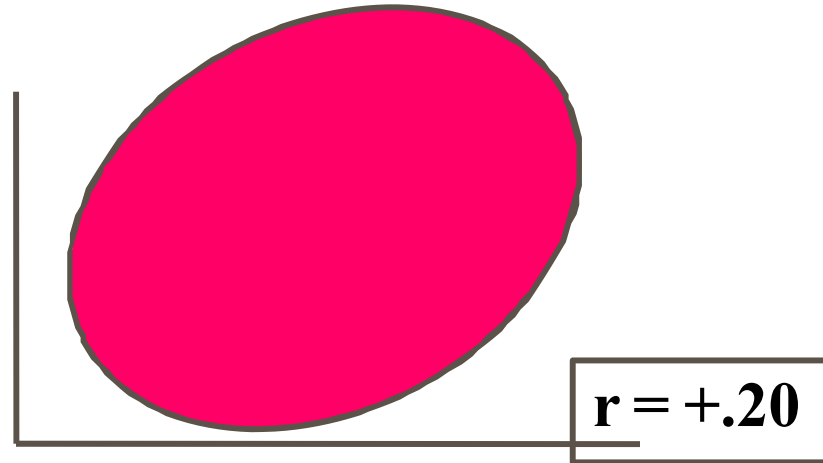
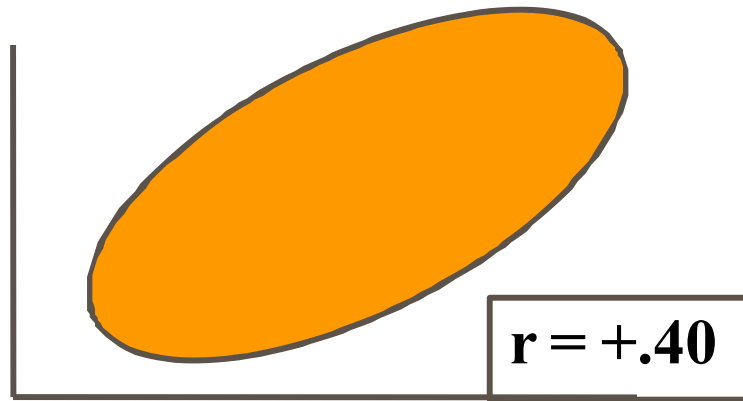
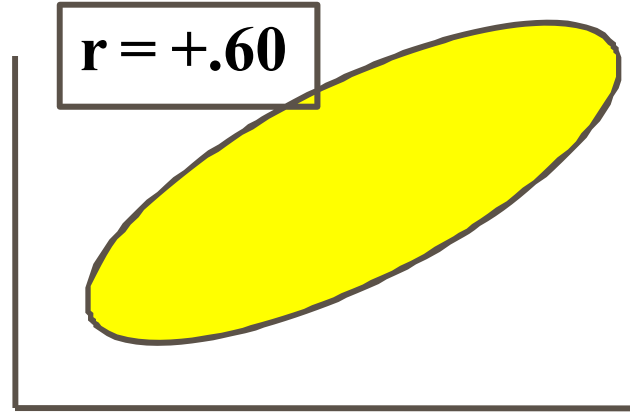
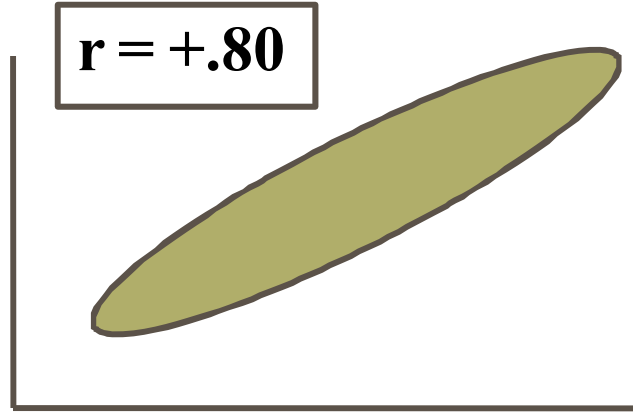
Strong positive

Weak Positive

■ Moderate Positive Correlation



Degree of correlation (r)



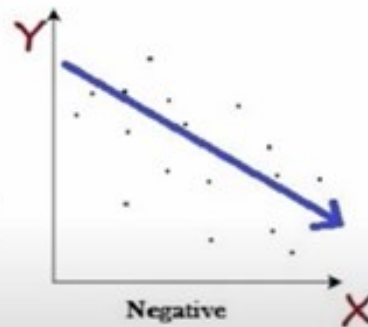
Negative or Inverse Correlation: If the values of the two variables deviate (moves) in the opposite direction, i.e., if the increase (decrease) in the values of one variable results, on an average, in a corresponding decrease (increase) in the values of the other variables.

Example:

- 1) Price and demand of a commodity.
- 2) Volume and pressure of a perfect gas. ($PV = \text{constant}$)

X	8	4	3	1
Y	8	10	15	25

X decreases; Y increases; ----- Negative Correlation



Degree of correlation:

Perfect Negative Correlation ($r = -1$): The variables move in opposite directions with a perfectly consistent linear relationship. As one variable increases, the other decreases in exact proportion.

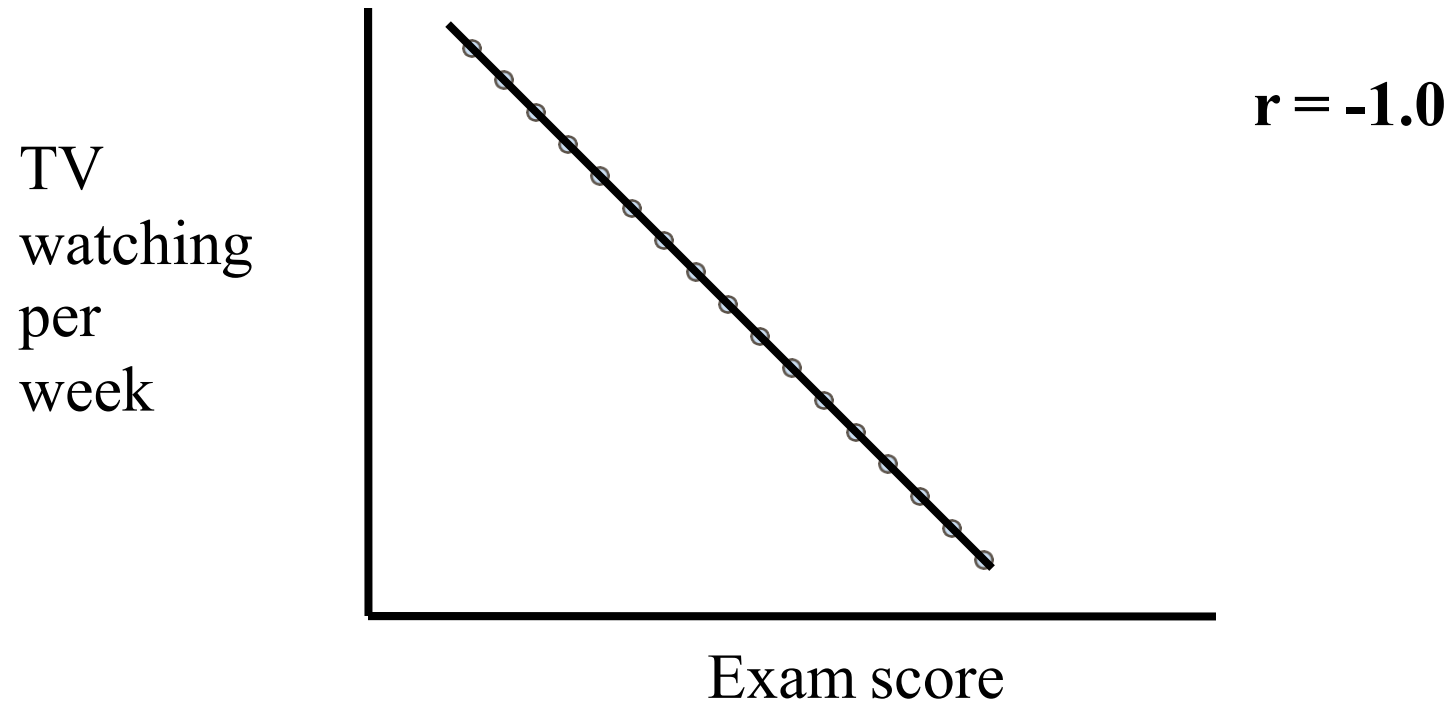
High Degree of Negative Correlation (strong) (r close to -0.7 to -0.99): There is a strong linear relationship, but the variables move in opposite directions. As one variable increases, the other decreases proportionally.

Moderate Negative Correlation (r around -0.5 to -0.7): The variables are negatively related, but the relationship is moderate. An increase in one variable is associated with a decrease in the other, though not perfectly.

Low Negative Correlation (weak) (r close to -0.2 to -0.5): There is a weak negative relationship. As one variable increases, the other tends to decrease slightly.

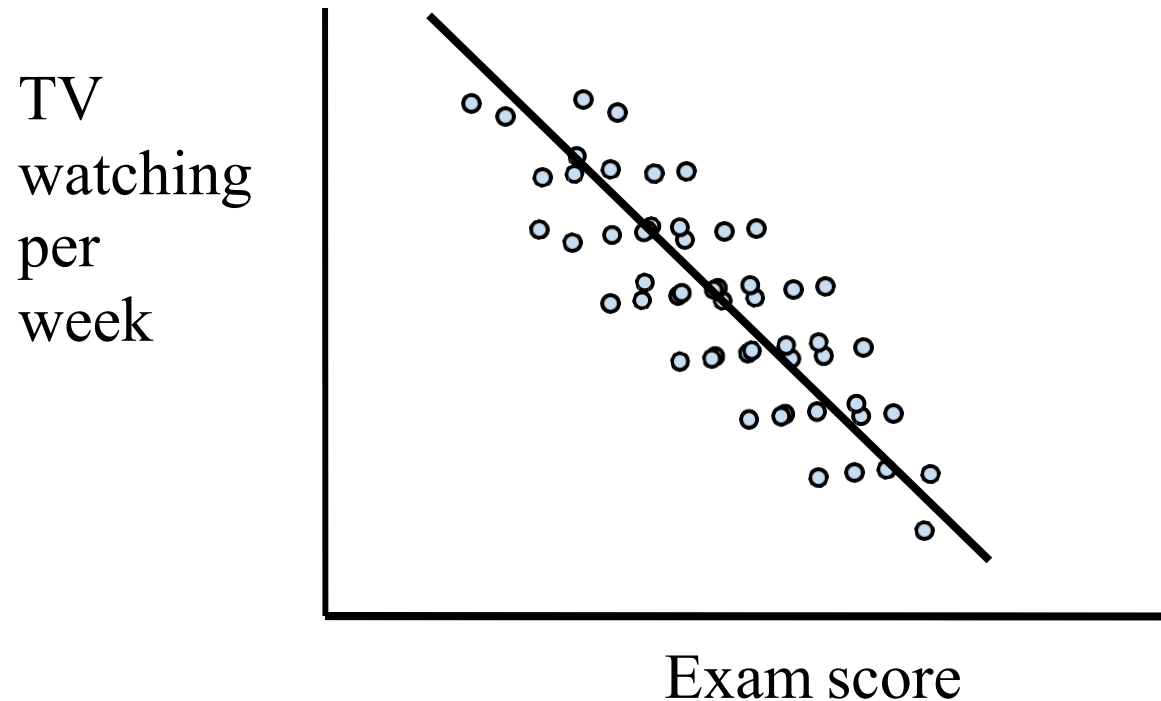
Degree of correlation

■ Perfect Negative Correlation



Degree of correlation

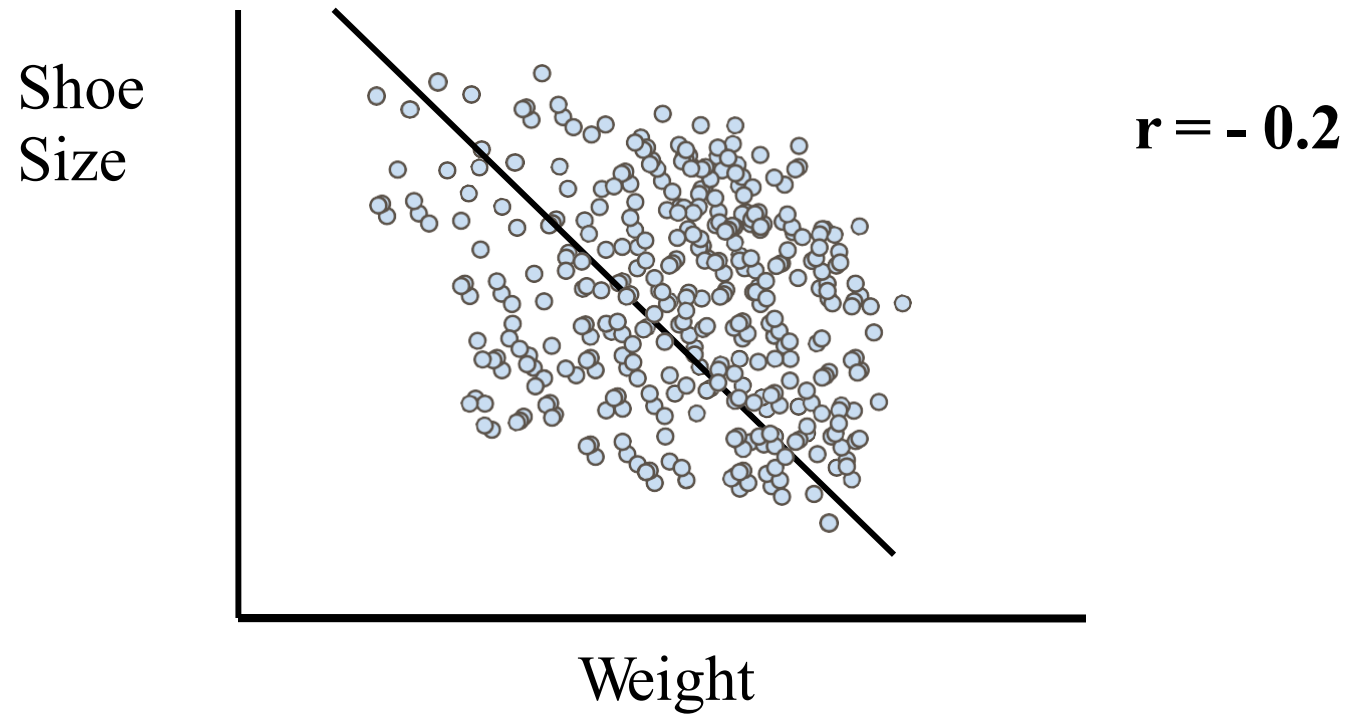
■ Moderate Negative Correlation



$$r = -0.60$$

Degree of correlation

■ Weak negative Correlation

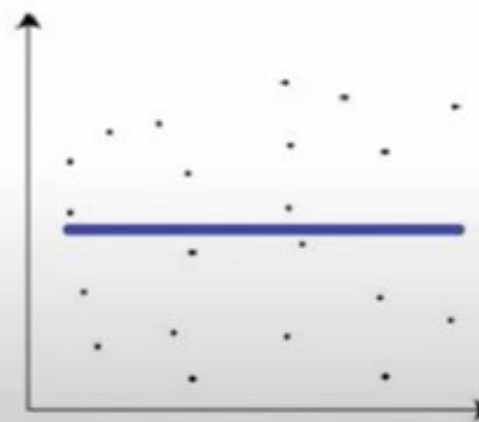


Weak/Zero correlation: It exists when one variable does not affect the other. For example, there is no correlation between the number of years of school a person has attended and the letters in his/her name.

Weak / no
correlation

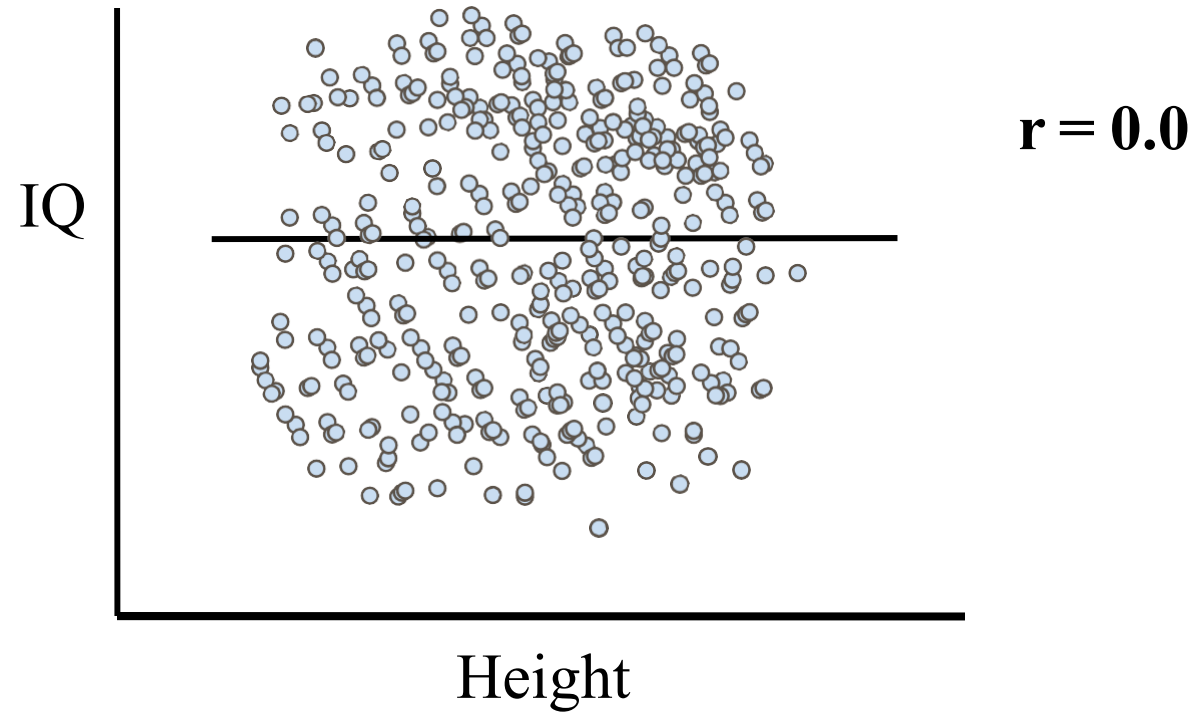


No correlation



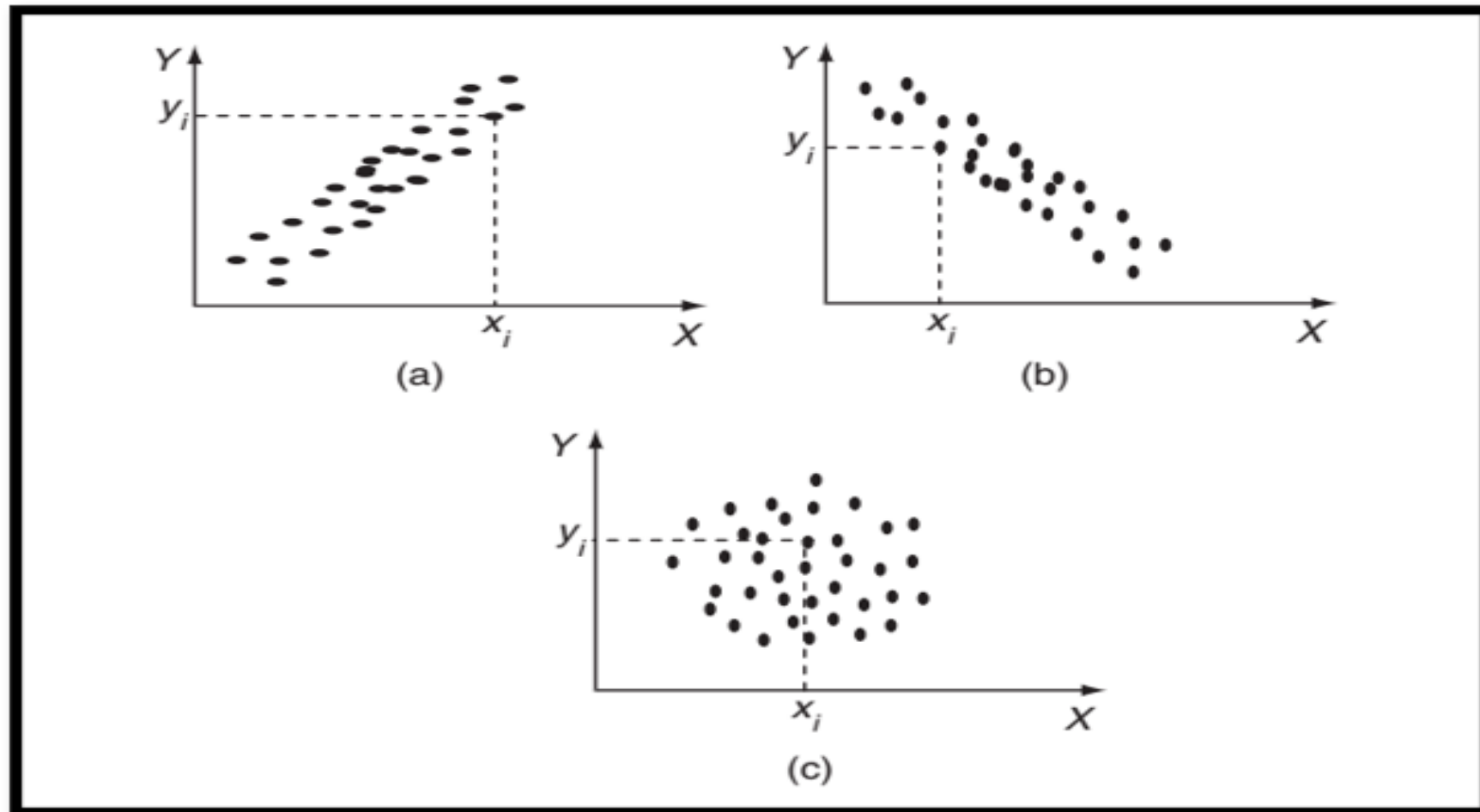
Degree of correlation

- **No Correlation (horizontal line)**



Question.

Identification of different types of correlation in the following diagram.

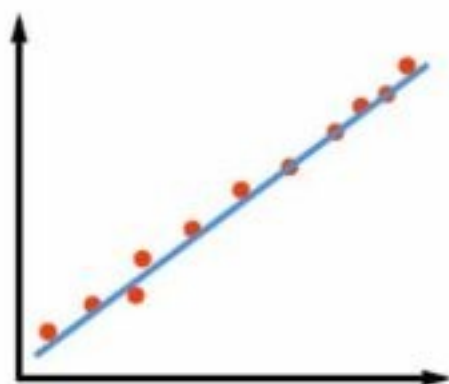


Answer –

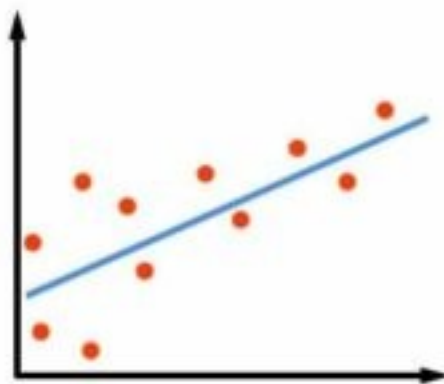
(a) Positive; (b) negative; (c) no correlations.

CORRELATION

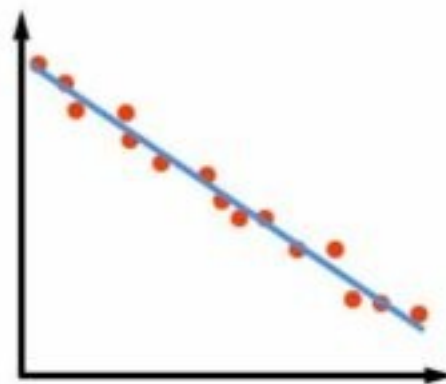
(INDICATES THE RELATIONSHIP BETWEEN TWO SETS OF DATA)



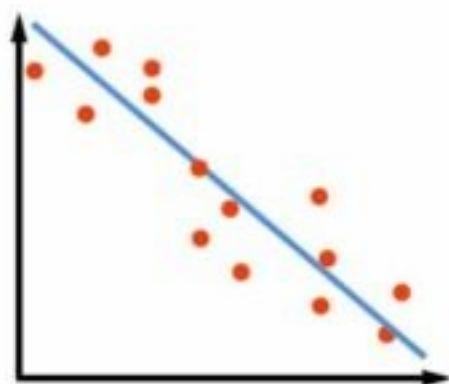
**STRONG POSITIVE
CORRELATION**



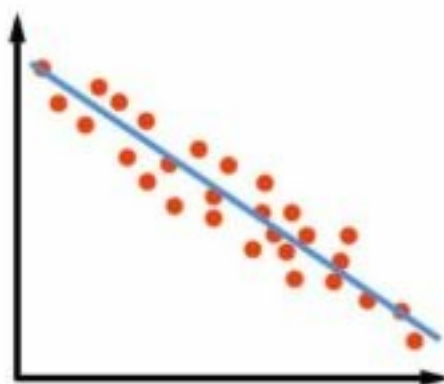
**WEAK POSITIVE
CORRELATION**



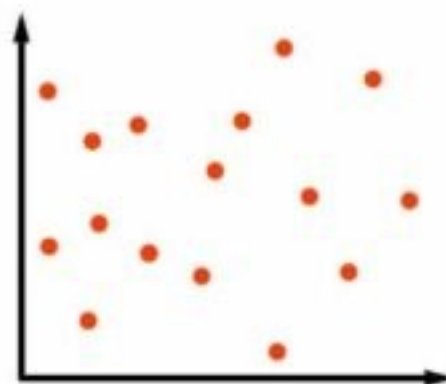
**STRONG NEGATIVE
CORRELATION**



**WEAK NEGATIVE
CORRELATION**



**MODERATE NEGATIVE
CORRELATION**



NO CORRELATION

Question.

Draw the scatter diagram and provide the conclusion.

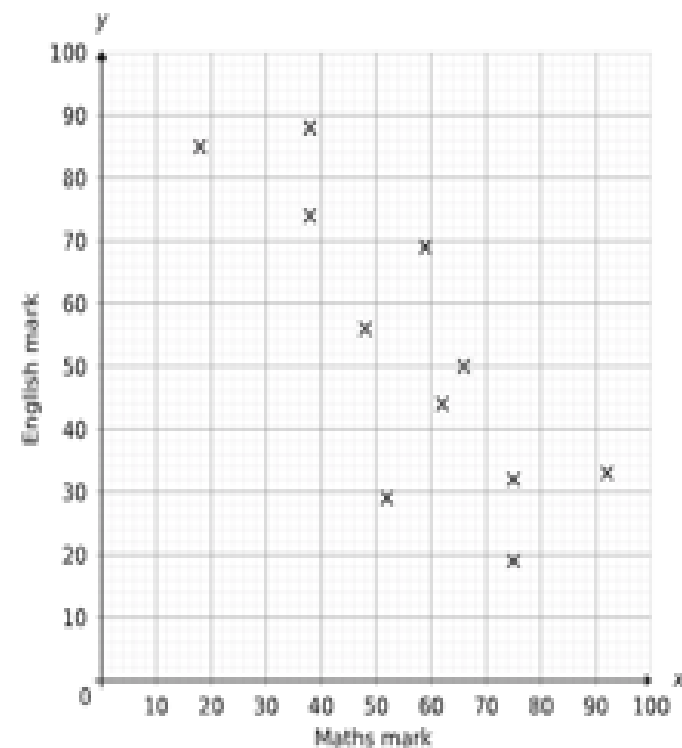
Below is a table of 11 student's scores out of 100 on their Maths and English tests. Plot a scatter graph from this data

Maths mark	38	62	18	75	38	59	66	92	52	75	48
English mark	74	44	85	19	88	69	50	33	29	32	56

Answer –

We will put the Maths mark on the x -axis and the English mark on the y -axis. Then, we plot each individual student's Maths mark against their English mark in the way that we normally plot coordinates.

Looking at this graph, we can see that, in general, as people's grades in Maths increase, their grades in English tend to decrease. So, there is **negative correlation** between Maths and English grades.



Question –

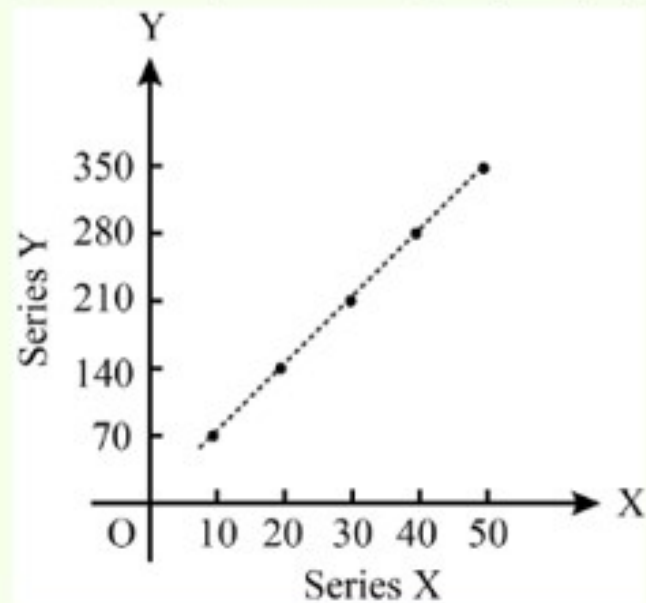
Make a scatter diagram for the following data and state the type of correlation between X and Y.

X	10	20	30	40	50
Y	70	140	210	280	350

Answer –

Solution

The scatter diagram is obtained by plotting the values of Series X on the X-axis and values of Series Y on the Y-axis. Plotting the values (10,70), (20,140), (50,350) on the graph paper, we get the scatter diagram:

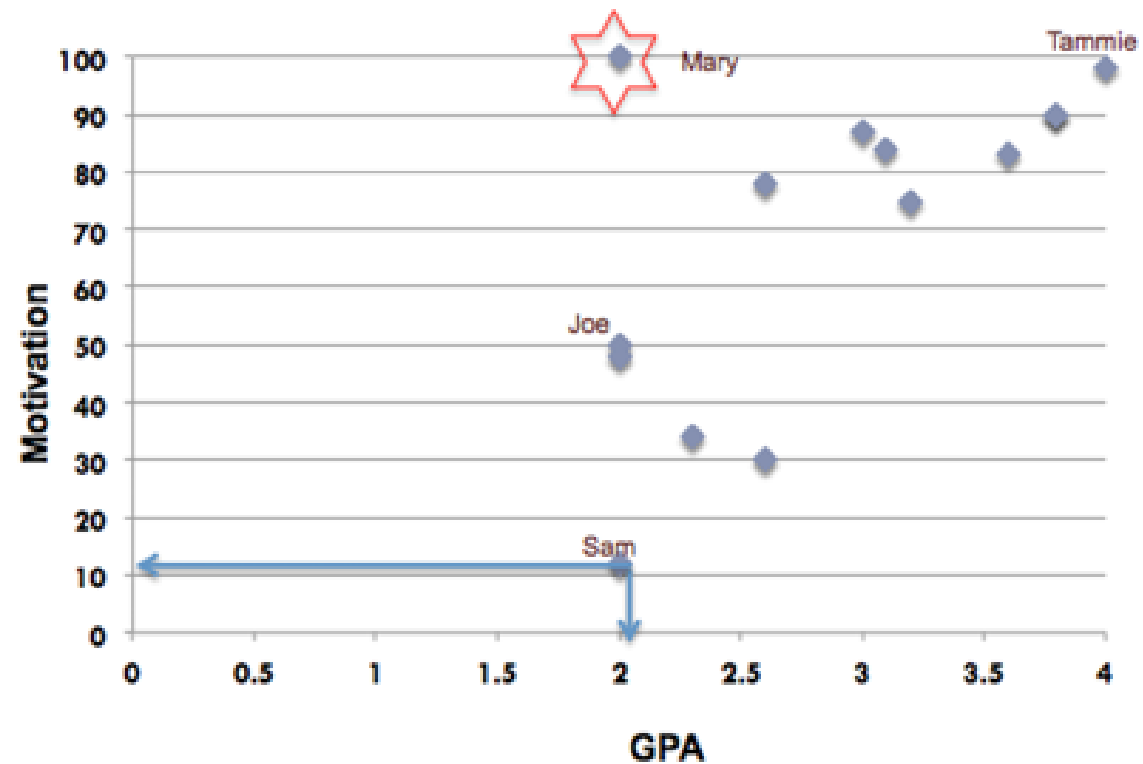


We can say that there is a perfect positive correlation between the values of Series X and Series Y.

Question –

What is the relationship between students' achievement motivation and GPA?

Student	Student GPA	Motivation
Joe	2.0	50
Lisa	2.0	48
Mary	2.0	100
Sam	2.0	12
Deana	2.3	34
Sarah	2.6	30
Jennifer	2.6	78
Gregory	3.0	87
Thomas	3.1	84
Cindy	3.2	75
Martha	3.6	83
Steve	3.8	90
Jamell	3.8	90
Tammie	4.0	98



Answer –

A **scatterplot** shows the relationship between two quantitative variables measured for the same individuals. The values of one variable appear on the horizontal axis, and the values of the other variable appear on the vertical axis. Each individual in the data appears as a point on the graph.

- Many research projects are **correlational studies** because they investigate the relationships that may exist between variables. Prior to investigating the relationship between two quantitative variables, it is always helpful to create a graphical representation that includes both of these variables. Such a graphical representation is called a **scatterplot**.

- In this example, the relationship between students' achievement motivation and their GPA is being investigated.
- The table on the left includes a small group of individuals for whom GPA and scores on a motivation scale have been recorded. GPAs can range from 0 to 4 and motivation scores in this example range from 0 to 100. Individuals in this table were ordered based on their GPA.
- Simply looking at the table shows that, in general, as GPA increases, motivation scores also increase.
- However, with a real set of data, which may have hundreds or even thousands of individuals, a pattern cannot be detected by simply looking at the numbers. Therefore, a very useful strategy is to represent the two variables graphically to illustrate the relationship between them.
- A graphical representation of individual scores on two variables is called a **scatterplot**.

- The image on the right is an example of a scatterplot and displays the data from the table on the left. GPA scores are displayed on the horizontal axis and motivation scores are displayed on the vertical axis.
 - Each dot on the scatterplot represents one individual from the data set. The location of each point on the graph depends on both the GPA and motivation scores. Individuals with higher GPAs are located further to the right and individuals with higher motivation scores are located higher up on the graph.
 - Sam, for example, has a GPA of 2 so his point is located at 2 on the right. He also has a motivation score of 12, so his point is located at 12 going up.
 - Scatterplots are not meant to be used in great detail because there are usually hundreds of individuals in a data set.
-
- The purpose of a scatterplot is to provide a general illustration of the relationship between the two variables.
 - In this example, in general, as GPA increases so does an individual's motivation score.
 - One of the students in this example does not seem to follow the general pattern: Mary. She is one of the students with the lowest GPA, but she has the maximum score on the motivation scale. This makes her an exception or an outlier.

Disadvantage of scatter diagram



Can not adopt the an exact degree of correlation

COEFFICIENT OF CORRELATION

Methods of Studying Correlation

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graph TD; A[Methods of Studying Correlation] --> B[1) Pearson product-moment correlation OR Karl Pearson's coefficient of correlation (Covariance method)]; A --> C[2) Rank Correlation coefficient method];
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- 1) Pearson product-moment correlation
OR
Karl Pearson's coefficient of correlation
(Covariance method).

- 2) Rank Correlation coefficient method

Karl Pearson's coefficient of correlation (Covariance method).

For two variables X and Y , the Pearson correlation coefficient is calculated as:

$$r = \frac{\text{Cov}(X, Y)}{\sigma_X \sigma_Y}$$

Where:

- $\text{Cov}(X, Y)$ is the covariance of X and Y .
- σ_X and σ_Y are the standard deviations of X and Y , respectively.

The Pearson correlation coefficient can also be expressed in terms of variances and covariances. For two random variables X and Y , the formula is:

$$r = \frac{\text{Cov}(X, Y)}{\sqrt{\text{Var}(X) \cdot \text{Var}(Y)}}$$

Where:

- $\text{Cov}(X, Y)$ is the covariance of X and Y .
- $\text{Var}(X)$ is the variance of X .
- $\text{Var}(Y)$ is the variance of Y .

Rules for finding Covariance

1st method

$$\text{cov}(X, Y) = E(XY) - E(X)E(Y)$$

$$\text{Var}(X) = E(X^2) - (E(X))^2$$

$$\text{Var}(Y) = E(Y^2) - (E(Y))^2$$

2nd method

$$\text{cov}(X, Y) = \frac{\sum (X - \bar{X})(Y - \bar{Y})}{n}$$

$$\text{Var}(X) = \frac{\sum (X - \bar{X})^2}{n}$$

$$\text{Var}(Y) = \frac{\sum (Y - \bar{Y})^2}{n}$$

Here are the Karl Pearson correlation formulas in the specific form you requested:

1. When the deviations are measured from the mean values:

$$r = \frac{\sum xy}{\sqrt{\sum x^2} \cdot \sqrt{\sum y^2}}$$

Where:

- $x = X - \bar{X}$
- $y = Y - \bar{Y}$
- $\sum xy$ is the sum of the products of the deviations of X and Y from their mean values.

2. If no assumed average is taken for X and Y series:

$$r = \frac{N \sum xy - \sum x \sum y}{\sqrt{N \sum x^2 - (\sum x)^2} \cdot \sqrt{N \sum y^2 - (\sum y)^2}}$$

Where:

- N is the number of data points.
- $\sum x$ and $\sum y$ are the sums of the individual values of X and Y .
- $\sum x^2$ and $\sum y^2$ are the sums of the squares of the values of X and Y .